a point $x \in X$ is neither an interior pt. nor an exterior pt. of A then it is called a boundary point of A. The set of all boundary points of A is called the boundary of A & it is denoted by ∂A or $bdy(A)$.

So

$$\partial A = X \setminus [\mathring{A} \cup ext(A)].$$

Note that

$$\mathring{A} \cap \partial A = \phi$$

$$ext(A) \cap \partial A = \phi$$

$$\mathring{A} \cap ext(A) = \phi.$$

Moreover

$$\mathring{A} \cup ext(A) \cup \partial A = X.$$

$$\begin{aligned} \partial(X \setminus A) &= X \setminus [(X \setminus A)^{\circ} \cup ext(X \setminus A)] \\ &= X \setminus [ext(A) \cup \dot{A}] \\ &= \partial A. \end{aligned}$$

Theorem: Let (X, τ) be a topological space & $A \subseteq X$. Then

(i) A is open $\iff A \cap \partial A = \phi$

(ii) A is closed $\iff \partial A \subseteq A$.

Proof:

(i) $\Longrightarrow$ A is open $\Rightarrow A = \mathring{A}$

$\Rightarrow A \cap \partial A = \mathring{A} \cap \partial A$

$= \phi$

$\Longleftarrow$ $A \cap \partial A = \phi$

$\Rightarrow A \subseteq X \setminus \partial A = \mathring{A} \cup ext(A)$

Since $A \cap ext(A) = \phi$, $A \subseteq \mathring{A}$

$\Longrightarrow A = \mathring{A}$, i.e. A is open.

(ii) A is closed $\iff X \setminus A \in \tau$

$\iff ext(A) = X \setminus A$

(i)

$\Longleftrightarrow ext(A) \cap \partial(X \setminus A) = \phi$

$\Longleftrightarrow X \setminus A \cap \partial A = \phi \iff \partial A \subseteq X \setminus (X \setminus A)$

$\Downarrow$ $\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\quad \| \; A.$

$\partial(X \setminus A) = \partial A$

Notes Let (X, τ) be a topological space & $A \subseteq X$.

(i) $ext(A)$ is open.

$$= (X \setminus A)^{\circ}$$

(ii) $\partial A = X \setminus [\overset{\circ}{A} \cup ext(A)]$

is closed

$\overset{\circ}{A}$: open; $ext(A)$: open; $\overset{\circ}{A} \cup ext(A)$: open

Example: Let $X = \{a, b, c, d, e\}$

& $\tau = \{\phi, \{a\}, \{c, d\}, \{a, c, d\}, \{b, c, d, e\}, X\}$. Consider $A = \{b, c, d\} \subseteq X$.

Determine $\overset{\circ}{A}$, $ext(A)$ & ∂A.

Soln

$\overset{\circ}{A} = \{c, d\}$

$X \setminus A = \{a, e\}$

$ext(A) = (X \setminus A)^{\circ} = \{a, e\}^{\circ} = \{a\}$

Hence

$\partial A = X \setminus [\overset{\circ}{A} \cup ext(A)]$

$= X \setminus \{a, c, d\} = \{b, e\}$.

Since $A \cap \partial A = \{b\} \neq \phi$, A is not open.

$\partial A \nsubseteq A$. Thus A is not closed.

Defnition: Let (X, τ) be a topological space & $A \subseteq X$. A point $x \in X$ is called a closure point of A iff x is not an exterior point of A. The set of all closure pt's of A is called the closure of A & it is denoted by $\overline{A}$, i.e. $\overline{A} = X \setminus ext(A)$.

Since $ext(A) = (X \setminus A)^\circ$ is open $\overline{A} = X \setminus ext(A)$ is always closed.

Notes:

1. $ext(A) = (X \setminus A)^\circ \subseteq X \setminus A$

$\Rightarrow$ $A \subseteq X \setminus ext(A) = \overline{A}$

2. $\mathring{A} \cup \partial A = \mathring{A} \cup [X \setminus [\mathring{A} \cup ext(A)]]$

$= \mathring{A} \cup [(X \setminus \mathring{A}) \cap (X \setminus ext(A))]$

$= (\mathring{A} \cup (X \setminus \mathring{A})) \cap (\mathring{A} \cup X \setminus ext(A))$

$= X \cap (\mathring{A} \cup \overline{A})$

$= \mathring{A} \cup \overline{A}$

$= \overline{A}$ $\quad (\mathring{A} \subseteq A \subseteq \overline{A})$

$\therefore \quad \overline{A} = \mathring{A} \cup \partial A.$

Proposition: Let (X, τ) be a topological space, $\mathcal{B}$ a base for τ & $A \subseteq X$. Then $x \in \overline{A} \iff B \cap A \neq \emptyset$ for every $B \in \mathcal{B}$ with $x \in B$.

Proof:

$\Longrightarrow$ Let $x \in \overline{A} = X \setminus ext(A)$. Then $x \notin ext(A) = (X \setminus A)^\circ$. Hence $U \not\subseteq X \setminus A$ for every $U \in \tau$ with $x \in U$. In particular for every $B \in \mathcal{B}$ with $x \in B$, $B \cap A \neq \emptyset$.

$\Leftarrow$ Assume that $B \cap A \neq \emptyset$ for every $B \in \mathcal{B}$ with $x \in B$. Suppose, to the contrary, that $x \notin \overline{A}$. Then $x \in ext(A) = (X \setminus A)^\circ$ & so there exists a $U \in \tau$ s.t. $x \in U \subseteq X \setminus A$. Since $\mathcal{B}$ is a base for τ there exists a $B \in \mathcal{B}$ s.t. $x \in B \subseteq U \subseteq X \setminus A$.

$\Rightarrow$ There exists a $B \in \mathcal{B}$ with $x \in B$ s.t. $B \cap A = \emptyset$. A contradiction.

$\therefore x \in \overline{A}$.

Note: If we replace the word "base" with the word "subbase" in the previous proposition the result may not be true:

Let $X = \{a, b, c, d\}$ & $\mathcal{S} = \{\{a, b\}, \{a, c\}, \{d\}\}$. Here $\mathcal{T} = \{\phi, \{a\}, \{d\}, \{a, b\}, \{a, c\}, \{a, d\}, \{a, b, d\}, \{a, c, d\}, \{a, b, c\}, X\}$. Let $A = \{b, c\} \subseteq X$. Let us check if $a \in \overline{A}$:

- Use the previous proposition:

$\mathcal{B} = \mathcal{S}^{\cap} = \{\phi, \{a\}, \{d\}, \{a, b\}, \{a, c\}, X\}$

Here $\{a\} \in \mathcal{B}$, $a \in \{a\}$. But $\{a\} \cap A = \phi$. Hence $a \notin \overline{A}$.

- Use $\mathcal{S}$ in the previous proposition:

All sets in $\mathcal{S}$ containing a are $\{a, b\}$ & $\{a, c\}$. Moreover $\{a, b\} \cap A = \{b\} \neq \phi$, $\{a, c\} \cap A = \{c\} \neq \phi$.